

A Hardware-Accelerated Post-Quantum Biometric Storage System

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Abstract—The impending advent of Cryptographically Relevant Quantum Computers (CRQCs) presents an existential danger to digital infrastructure based upon classical asymmetric cryptosystems. Shor's algorithm allows a suitably large quantum computer to compute both prime factorization and the discrete logarithm in polynomial time, thus making RSA and ECC systems unusable going forward. Biometric identity management systems are uniquely susceptible to attack using such techniques due to the immutability of the underlying templates. This paper describes the design, mathematics, software implementation, and empirical analysis of a Quantum-Resistant Biometric Data Vault built around a novel Key Encapsulation / Data Encapsulation (KEM-DEM) design combining three cryptographic primitives: NIST-standardized FIPS 203 (ML-KEM-768) quantum-resistance KEM, FIPS 204 (ML-DSA-44) lattice-based signature generation, and AES-256-GCM symmetric bulk encryption. The prototype solution uses hardware-accelerated AVX2/AVX-512 support provided by the open source liboqs cryptographic library. Experiments performed over 1,000 iterations show sub-millisecond latencies per function call: 0.0234 ms for KEM function calls and 0.0925 ms for signatures, yielding a complete store operation at ~ 0.1577 ms. On resource-constrained ARM CortexM0+ edge devices, ML-KEM-768 shows a 94% improvement in energy efficiency over classical ECDH P-256 scheme. A regression analysis additionally highlights an architectural limitation: ML-DSA-44 produces 2,420-byte signatures exceeding the maximum Ethernet MTU of 1,500 bytes, thereby requiring IP fragmentation and significantly increasing probability of TCP Retransmission Timeout (RTO) events to 51% in high loss LEO satellite links.

Keywords—Post-Quantum Cryptography, Biometric Security, ML-KEM, ML-DSA, FIPS 203, FIPS 204, KEM-DEM Architecture, Lattice-Based Cryptography, Hardware Acceleration, IoT Edge Security

I. INTRODUCTION

The underlying foundation of modern cryptography, including the security of communications, finances, and identity, relies heavily on the computational complexity of certain mathematical problems. RSA cryptosystems and elliptic curve cryptography rely on the computational complexity of prime factorization and the discrete logarithm problem, respectively. For many decades, classical asymmetric algorithms were used to secure TLS handshakes, PKI certificate chains, secure emails, and digital banking transactions.

However, the rapid advancements in quantum information science are about to change the entire game. In 1994, Peter Shor proposed that quantum computers with sufficient qubits would be able to use superposition and entanglement to quickly perform prime factorization and discrete logarithms in polynomial time [9]. With the invention of Cryptographically Relevant Quantum Computer (CRQC), all current public key encryption techniques will become entirely ineffective. The international community took decisive action by launching the NIST Post-Quantum Cryptography Standardization project in 2016, which concluded with the publication of FIPS 203, FIPS 204, and FIPS 205 in 2024. [2][3][4].

Fingerprint minutiae maps, facial geometry encodings, iris pattern templates, palm vein structure information, voice print templates and behavioural biometrics including gait analysis belong to another category of identifiers. Biometric credentials rely on a template built using attributes about the owner of the credential which cannot be changed. Indeed, no person will ever be able to change fingerprints, iris geometry, nor palm vein configuration. A compromised biometric identifier means lifelong impersonation exposure for the affected person.

According to the NSA and NIST, the advent of CRQC in ten to fifteen years is a realistic planning scenario. The implementation of quantum-resistant cryptography should therefore take place immediately due to the threat of Harvest Now, Decrypt Later (HNDL) attacks.

A. The Distinct Susceptibility of Biometric Identity Systems

Whereas the quantum threat affects the entire cryptosystem, biometric credentials management would suffer from an unprecedentedly irreversible impact. Classical security mechanisms assume that when one cryptographic key is compromised, another one can be immediately generated. However, if a biometric credential becomes compromised, there is no way to remedy the situation. Fingerprint minutia, facial geometry, iris patterns, palm vein networks, and gait analysis signatures represent physiological features that cannot be changed or replaced once obtained.

Therefore, such a property makes an attack on HNDLs particularly detrimental. Modern Advanced Persistent Threat (APT) actors intercept and store all biometric credentials sent through authentication applications programming interfaces (APIs), border control databases, and healthcare identity management systems. Once CRQC technology becomes available, these stored biometric templates will become permanently exposed, allowing for identity theft, illegal crossing of borders, and extensive surveillance.

B. Research Gap and Contributions

However, a survey of the current body of literature does not show any prior biometric vault architecture design meeting all four criteria discussed: (1) full protection against CRQC attack; (2) computational efficiency necessary to achieve high processing speed during the identity verification; (3) low energy consumption, enabling its usage on IoT edge devices equipped with batteries; and (4) usage of only standardized government frameworks rather than preliminary theoretical models. While the first condition is met by the Fully Homomorphic Encryption (FHE), the latter two are not, as the latencies in this approach are more than 1.24 s, and the memory is more than 20 MB [1].

This work contributes to the existing research through four main contributions: (i) a formal description of a hybrid KEM-DEM biometric vault architecture using FIPS 203, FIPS 204, and AES-256-GCM algorithms in a Sign-then-Encrypt model; (ii) performance analysis via experiments performed

over 1,000 iterations of the implemented solution accelerated with AVX2/AVX-512 instructions; (iii) energy consumption profiling on ARM Cortex-M0+ microcontrollers; and (iv) a regression analysis of the impacts of signature size growth on IP packet fragmentation and TCP RTOs.

II. METHODOLOGY

A. Biometric Data Pipeline

For testing the cryptographical vault prototype, biometric authentication is simulated using high-resolution digital fingerprint template samples, which are modelled to simulate the real-world outputs of optical or capacitive sensor types. The modern approach to storing biometric information does not involve storing unprocessed sensor data; rather, unique biometric characteristics are extracted using sophisticated algorithms to be represented mathematically. Biometrics' fingerprint analysis is achieved using minutiae extraction. These spatial locations along with ridge angle and type make up a multidimensional vector that is analysed using CNNs for ridge segmentation and SVM for minutiae extraction. The resulting feature vector is encoded in a fixed length 512-byte binary format and normalized, creating uniform input that can be used in ML-DSA internal hash function operations and AES-256-GCM block cipher padding.

B. Hybrid KEM-DEM Architecture

This data vault relies on a mixed approach of a Key Encapsulation Mechanism and a Data Encapsulation Mechanism (KEM-DEM). The first asymmetric KEM module establishes a temporary symmetric secret key via a non-secure channel. Afterward, the second symmetric DEM module encrypts the payload using this key.

Four distinct stages form the protocol pipeline. In the KEM encapsulation stage, ML-KEM-768 uses the server's public key to generate a new ephemeral secret, resulting in the formation of a 1,088-byte ciphertext and a 32-byte shared secret that cannot be extracted without knowing the server's private key. The second stage is the payload encryption stage, where the generated shared secret is used to instantiate an AES-256-GCM cipher algorithm. A nonce with 96-bit length, produced by the CSPRNG algorithm, is used to encrypt the normalized payload and its 44-bit digital signature, ML-DSA-44, into ciphertext and an authentication tag with 128 bits.

C. Mathematical Foundations: Module-LWE

Both FIPS 203 and FIPS 204 derive their post-quantum security guarantees from the hardness of the Module Learning with Errors (M-LWE) problem introduced by Regev in 2005 [8]. In the standard LWE construction, an error vector e drawn from a truncated Gaussian distribution transforms the tractable linear system $b = A \cdot s \pmod{q}$ into $b = A \cdot s + e \pmod{q}$. The injected noise renders the distribution of b computationally indistinguishable from a uniform random vector to any polynomial-time adversary without knowledge of s .

The Module variant (M-LWE) extends this construction to polynomials from the ring $R_q = \mathbb{Z}_q[X]/(X^n + 1)$, with $n = 256$ and q a designated prime modulus for ML-KEM and ML-DSA. This ring structure enables the NTT optimization described below while preserving security hardness reductions to the Shortest Vector Problem (SVP) in high-dimensional lattices. Quantum computers cannot efficiently solve M-LWE because Shor's algorithm exploits periodic structure in modular exponentiation—a property absent from the aperiodic geometry of lattice problems. ML-KEM-768

targets NIST Security Level 3, requiring at least 2^{178} classical bit operations to break.

D. NTT Optimization and Hardware Acceleration

Polynomial multiplication is the primary operation in lattice-based cryptography. Naive schoolbook multiplication of polynomials of degree $(n-1)$ has complexity $O(n^2)$; thus, for $n = 256$, each polynomial multiplication involves 65,536 operations. The Number Theoretic Transform (NTT), which is an implementation of the Fast Fourier Transform in finite fields, reduces the complexity of polynomial multiplication to $O(n \log n)$, providing nearly 100 times speedup when $n = 256$. This speed-up comes through using a primitive $2n$ -th root of unity, such that $\omega^n \equiv -1 \pmod{q}$, allowing negacyclic convolution and obtaining products modulo $X^n + 1$ with exact integer arithmetic and no errors from rounding.

The Open Quantum Safe (liboqs) cryptographic library utilizes deep Intel AVX2 and AVX-512 acceleration in its implementation of these computations. AVX-512 offers 32 independent 512-bit vector registers (zmm registers) capable of performing 16 32-bit multiplications simultaneously. Published benchmarking results comparing ML-DSA implementations with optimized AVX-512 and AVX2 versions show keygen performance improvement by $2.20\times$ and signature rate improvement by $2.36\times$ [5].

E. FIPS 203 and FIPS 204 Parameter Specifications

ML-KEM-768 employs a 3×3 module matrix of degree-255 polynomials over $\mathbb{Z}_q[X]/(X^{256} + 1)$. Rather than transmitting the full public matrix A , the public key stores only a 32-byte seed ρ from which A is deterministically expanded via SHAKE-128, compressing the public key to 1,184 bytes. ML-DSA-44 is built on the Fiat-Shamir-with-Aborts paradigm over Module-LWE and Module-Short Integer Solution (M-SIS) hardness assumptions at NIST Security Level 2. Its rejection sampling mechanism probabilistically discards signing attempts whose output signatures could leak secret key structure, introducing geometric distribution in signing latency. Key parameters are summarized in Table I.

TABLE I. ML-DSA-44 Parameter Specifications (FIPS 204)

Parameter	Symbol	Value	Significance
Matrix Dimensions	k, l	4×4	Defines module rank
Coefficient Modulus	q	8,380,417	Prime enabling NTT
Private Key Range	η	2	Secret key bound
Public Key Size	—	1,312 bytes	Storage requirement
Signature Size	—	2,420 bytes	Exceeds 1,500-byte MTU

III. RESULTS AND ANALYSIS

A. Computational Latency Benchmarks

Empirical benchmarking was conducted on a reference server platform running Ubuntu 22.04 LTS, equipped with an Intel AMD Ryzen 4600H processor, 16 GB DDR4 RAM, and NVMe SSD storage. Across 1,000 independent execution iterations with CPU frequency scaling disabled and cache warmup completed prior to measurement collection, the hardware-accelerated hybrid KEM-DEM vault demonstrates comprehensively sub-millisecond operation latencies, as presented in Table II.

TABLE II. Empirical Execution Latencies (1,000 Iterations, AVX-512)

Cryptographic Operation	Algorithm	Avg. Latency
Key Generation	ML-KEM-768	0.0198 ms
Key Encapsulation	ML-KEM-768	0.0234 ms
Signature Generation	ML-DSA-44	0.0925 ms
Signature Verification	ML-DSA-44	~0.1400 ms
AES-256-GCM Enc. (512B)	FIPS 197	0.0018 ms
Full Vault Store Operation	Composite KEM-DEM	~0.1577 ms

The full vault store operation—comprising ML-DSA signing, ML-KEM encapsulation, and AES-256-GCM encryption—completes in approximately 0.1577 ms, enabling a single processor core to handle approximately 6,340 biometric storage operations per second. By comparison, RSA-2048 key generation requires approximately 2.3 ms and RSA-2048 signing approximately 0.8 ms on the same hardware—both substantially slower than the complete post-quantum vault operation. The findings unequivocally debunk the general presumption that post-quantum cryptography entails an insurmountable computational burden. Such a belief stems from early-stage, poorly optimized research implementations that cannot be applied to practical AVX-512-optimized systems adhering to the final NIST protocols.

B. Energy Efficiency on Constrained Edge Architectures

An energy profiling analysis of the ARM Cortex-M0+processor was performed, a typical ultra-low power 32-bit embedded CPU used in IoT fingerprint recognition modules, biometric passports, and wearables for individual identification. According to results shown in Table III, the time required by ML-KEM-768 to complete an entire KEX is 56.62 ms and consumes just 4.50 mJ; however, ECDH P-256 requires more than 150 ms and uses 48.87 mJ—consuming 89% less energy. RSA-2048 requires over 250 ms and uses 279.80 mJ of energy, which is 62.2 times higher than ML-KEM-768. When considering battery operational life with a single CR2032 coin cell, RSA-2048 can perform approximately 32000 cycles. Vs 20,00,000 for ML-KEM-768.

TABLE III. Energy Profiles on ARM Cortex-M0+ Microarchitecture

Cryptographic Operation	Completion Time	Energy Consumed
RSA-2048 (Baseline)	> 250.0 ms	279.80 mJ
ECDH P-256 (Classical)	> 150.0 ms	48.87 mJ
ML-KEM-768 (post-Quantum)	56.62 ms	4.50 mJ
ML-KEM-512 (post-Quantum)	35.70 ms	2.83 mJ

An important limitation exists for the performance of post-quantum digital signatures on constrained hardware. Due to the use of rejection sampling by ML-DSA, the latency for performing digital signatures exhibits geometric distribution. The maximum signing latency occurs at the 99th percentile with latency values reaching 1,125 ms, while power consumption peaks at a few hundreds of millijoules. In such cases, it is advised to utilize the FN-DSA (Falcon) algorithm.

C. MTU Bottleneck and IP Packet Fragmentation

The constraints on the mathematical parameters defined by the FIPS 204 mathematical parameters result in a mandatory signature size of precisely 2,420 bytes for ML-DSA-44—a mathematical requirement imposed by fulfilling the NIST security level 2 criteria and not an implementation limitation. The MTU universally set for Ethernet packets according to RFC 791, published in 1981, is 1,500 bytes [6]. As such, the 2,420-byte ML-DSA-44 signature is 920 bytes larger than the MTU (61.3%) and requires fragmentation at the IP layer into at least two segments.

IP fragmentation introduces compounding failure risk: if any single fragment is lost, the entire datagram must be retransmitted from the beginning. Regression modelling of fragmented ML-DSA-44 signature transmission over a simulated LEO satellite link with a 30% packet error rate forecasts a 51% probability of catastrophic TCP Retransmission Timeouts (RTOs) exceeding 1,000 ms per authentication handshake. The Falcon (FN-DSA) scheme, producing 858-byte signatures that fit within a single MTU packet, yields only a 30% RTO probability under identical conditions. Results are summarized in Table IV.

TABLE IV. Regression Analysis: Signature Size vs. RTO Probability (30% Loss)

Algorithm	Size	MTU Compatible?	RTO Probability
ECDSA P-256	64 B	Yes (Single Packet)	< 1% (Baseline)
Falcon (FN-DSA)	858 B	Yes (Single Packet)	30%
ML-DSA-44	2,420 B	No (Fragmentation)	51%

In LEO satellite environments with conservative Initial Congestion Window configurations of 1 segment, the mandatory 1-RTT penalty from fragmentation introduces a baseline latency of 20–40 ms per authentication handshake. The actual end-to-end authentication latency of ML-DSA-44 in network-constrained deployments may therefore be 200 to 500 times greater than its raw CPU-level signing latency suggests.

IV. DISCUSSION

A. Computational Performance Readiness

The benchmark results confirm that the hardware-accelerated hybrid KEM-DEM architecture is production-ready for server-side biometric vault deployments. The full vault operation latency of approximately 0.1577 ms is well within the 500 ms to 2 s end-to-end authentication response time envelope required by high-throughput identity verification systems. An 8-core server can process over 50,000 biometric store operations per second without architectural optimization, substantially exceeding the throughput requirements of most large-scale identity management deployments.

The 94% saving on energy in comparison with ECDH P-256 is an unusual alignment of benefits in terms of security and performance. Architects used to be faced with the dilemma of opting for a secure but non-performant ECDH in exchange for better battery life. Now, with ML-KEM-768, this dilemma no longer applies.

B. Network Fragmentation and Mitigation Strategies

The problem posed by MTU fragmentation constitutes the most critical practical impediment for implementing ML-DSA-44 in network-facing authentication mechanisms. NIST standards have put more emphasis on cryptographic

and mathematical correctness rather than on network communication considerations. Network engineers are left to bridge this gap themselves. Path MTU Discovery (PMTUD) defined in RFC 1191 [7] is a way for a TCP session to discover path MTU using fragmentation needed ICMP messages, which however, are commonly blocked by corporate and governmental firewalls. Therefore, it cannot be considered a comprehensive solution.

Algorithm substitution can be used as an effective means of mitigating MTU fragmentation problems. For real-time authentication, Falcon (FN-DSA-512) offers Level 1 quantum resistance and uses 858-byte signatures (no fragmentation). This can be used for online operations where network communication delay does not matter much, while ML-DSA-44 could be utilized for offline or batch processing tasks where latency is less critical. In cases when ML-DSA-44 must be used due to its NIST Level 2 quantum protection requirement in a real-time environment, application-level or QUIC transport layer segmentation can be considered.

C. Threat Model and Residual Risks

The overall security analysis needs to consider more threats other than the hardness of the cryptographic primitives used. Such kinds of threats include attacks on the physical implementation of encryption schemes through side-channel analysis such as timing attacks, simple power analysis, differential power analysis, electromagnetic emission attacks, and cache-timing attacks. Liboqs offers protection against cache and timing analysis threats due to its constant time implementation, while complete side-channel protection on physical devices may require specialized hardware solutions.

The enrolment process in biometrics is also a crucial surface from which security threats can emerge. Compromise of the enrolment devices through attacks such as printing fingerprints, facial image resolution attacks, and malicious enrolment terminals will make the encryption strength irrelevant since no security assurance would be provided after that point. Liveness checks, authentic enrolment devices, and biometric fusion are some of the security considerations to make during the implementation phase. Ultimately, the security of the vault is dependent on the secrecy of the server private key of the ML-KEM scheme.

V. CONCLUSION

It can be seen that the architecture of the Quantum-Resistant Biometric Data Vault is not only mature but also strategically essential. As previously outlined, the hybrid KEM-DEM architecture succeeds in meeting all four research requirements identified in the research gap analysis: quantum resistance due to the use of standardized FIPS 203 and FIPS 204 primitives; real-time processing capability verified through sub-millisecond latency and over 6,000 computations per second per core; a 94% energy efficiency improvement compared to ECDH for constrained IoT devices; and an exclusive commitment to government-approved NIST standards. Moreover, the deliberate architectural decision to isolate the computationally heavy post-quantum key exchange and signatures from the bulk symmetric AES-256-GCM encryption, alongside the use of NTT multiplication and AVX-512 instructions, succeeded in mitigating the computational bottleneck issues found in prior post-quantum cryptography research.

In terms of architectural constraints, the only critical factor is the necessity for IP packet fragmentation resulting from the ML-DSA-44 signature having a size of 2,420 bytes, which

exceeds the 1,500-byte standard MTU limit—this being a mathematical constraint of NIST Security Level 2 parameters rather than a limitation of the actual architecture itself. With regards to applications in the high-latency, unreliable LEO satellite network with its higher packet loss rate, the fragmentation penalty can result in up to three orders of magnitude higher end-to-end latency compared to the CPU operation time. Potential future research could include hybrid protocols allowing for network characterization to switch between Falcon and ML-DSA-44; QUIC transport encapsulation to avoid head-of-line blocking; and smaller intermediate ML-DSA parameter set.

Entities handling sensitive biometric databases—national ID providers, border protection services, health care systems, banks, and enterprise access control managers—have no choice but to make adoption of NIST PQ cryptography their number one operational concern. The Harvest Now, Decrypt Later strategy is real, and state actors with the capability are systematically storing encrypted biometric data for eventual decryption using CRQC. The quantum deadline is not some far-off date in the future, but a concrete deadline that must be met. And once that deadline passes, it's too late.

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